Bayesian Statistics III: Bayesian Linear Regression

ECON 8310: Business Forecasting — Lecture 12

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August 24, 2026

Lecture Outline

1 Bayesian Linear Regression

2 Interpreting Posterior Coefficient Distributions

3 Directed Acyclic Graphs and Causal Structure

4 Conditional Analysis and Group-Specific Models

5 Course Synthesis and Next Steps

Bayesian Linear Regression

OLS gives a single coefficient estimate. Bayesian regression gives a full distribution over every coefficient, quantifying exactly how uncertain we are about each effect.

OLS vs. Bayesian Regression: What Changes?

OLS (Lectures 2 and 8):

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{y}$$

- Single point estimate $\hat{\boldsymbol{\beta}}$
- Standard errors derived from asymptotic theory
- Confidence intervals rely on large-sample normality
- Regularization (Ridge, LASSO) is an ad-hoc fix

Bayesian regression:

$$p(\boldsymbol{\beta} \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta})$$

- Full posterior distribution for each β_j
- Credible intervals are exact probability statements
- Regularization = informative prior (Normal $\equiv$ Ridge; Laplace $\equiv$ LASSO)
- Works even with small samples

Bayesian regression answers questions OLS cannot: "What is the probability that advertising elasticity exceeds 0.5?" or "Did price sensitivity increase or decrease?"

Bayesian Linear Regression in PyMC

Retail Sales Regression

```
with pm.Model() as blr:
intercept = pm.Normal('intercept',
mu=y.mean(), sigma=y.std())
beta = pm.Normal('beta',
mu=0, sigma=1, shape=X.shape[1])
sigma = pm.HalfNormal('sigma', y.std())
mu = intercept + X @ beta
pm.Normal('y_obs', mu, sigma, observed=y)
trace = pm.sample(2000)
pm.plot_posterior(trace, var_names=['beta'])
```

Standardize predictors first:

- Center and scale: `X_std = (X - X.mean()) / X.std()`
- With standardized inputs, $\mathcal{N}(0, 1)$ priors are weakly informative
- Coefficients $=$ "effect of 1-SD increase in x_j on y"

A Normal prior on β is equivalent to Ridge regularization. A Laplace prior is equivalent to LASSO. Bayesian inference subsumes regularization.

Interpreting Posterior Coefficient Distributions

Credible intervals, probability of direction, and conditional effect analysis.

Reading Posterior Distributions

For each β_j, the posterior gives:

Posterior mean
: Best single estimate

Posterior median
: More robust if distribution is skewed

HDI
: Highest Density Interval: shortest interval containing $X\%$ of the probability mass

Prob. of direction
: $P(\beta_j > 0)$ or $P(\beta_j < 0)$ — more natural than a p-value

ROPE
: Region Of Practical Equivalence: fraction of the posterior in $|\beta_j| < \delta$ ("negligible effect"). High ROPE mass = statistically present but practically irrelevant.

Business Example

Posterior for advertising elasticity β_{ads}:

- Posterior mean: 0.32
- 94% HDI: [0.18, 0.47]
- $P(\beta_{\text{ads}} > 0) = 0.998$

Interpretation: "99.8% probability of a positive effect. A 10% increase in ads raises sales by 3.2% (credible range: 1.8%–4.7%)."

DAGs and Causal Structure

A Directed Acyclic Graph makes the assumed causal structure explicit, helping decide which variables to include in a model.

Directed Acyclic Graphs (DAGs)

A **DAG** is a graph where:

- **Nodes** = variables
- **Directed arrows** = causal relationships ($A \to B$: A causes B)
- **Acyclic** = no variable causes itself

How to build: For each variable pair, ask "Could A directly cause B?" Draw an arrow if yes; direction = causation, not correlation.

Why use DAGs?

- Makes causal structure explicit before modeling
- Identifies **confounders** (must control for)
- Identifies **colliders** (must NOT control for)
- Clarifies which coefficients are causal vs. associational

Retail Sales DAG

- Advertising $\to$ Sales
- Price $\to$ Sales
- Season $\to$ Advertising
- Season $\to$ Sales

Season is a confounder: omitting it causes the advertising coefficient to absorb seasonal variation — biased upward.

Conditional Analysis and Group-Specific Models

Use posterior samples to ask conditional questions: what happens to the forecast if this variable changes?

Scenario Analysis with Posterior Samples

Procedure:

1. Draw S posterior samples $\boldsymbol{\beta}^{(1)}, \ldots, \boldsymbol{\beta}^{(S)}$
2. For scenario $\mathbf{x}^*$, compute $\hat{y}^{(s)} = \mathbf{x}^{*\top}\boldsymbol{\beta}^{(s)}$
3. The distribution of $\hat{y}^{(s)}$ is the **posterior predictive distribution**

"What if we raise price by 5%?"

```
x_new = x_base.copy()
x_new['price'] *= 1.05
b = trace.posterior['beta'].values
y_draws = b @ x_new
print(f"E[sales]: {y_draws.mean():.0f}")
print(f"95% CI: {np.percentile(y_draws,[2.5,97.5])}")
```

Group-specific models:

- Hierarchical structure (L11) gives **store-specific** posteriors
- Scenario effects differ by store: high-elasticity vs. low-elasticity
- Posterior uncertainty reveals which stores drive total revenue risk

Decision analysis: choose the action (price, promo, budget) that maximizes expected profit given the full posterior over outcomes.

Course Synthesis and Next Steps

From classical time series to Bayesian hierarchical models: a complete forecasting toolkit.

Course Method Map

Method	Data	Uncertainty	Interpretable	Strengths
ETS / ARIMA (L01–02)	Univariate TS	CI	High	Fast, automatic
GAM / Prophet (L03)	TS + predictors	Limited	High	Nonlinear + interpretable
Decision Tree (L04)	Tabular	No	High	Rule-based explanation
Random Forest (L05)	Tabular	OOB CI	Medium	Robust, minimal tuning
XGBoost (L06)	Tabular	No	Low	Best tabular accuracy
FFN / 1D CNN (L07–08)	Tabular / TS	No	Low	Smooth functions, fast
LSTM / Transformer (L09)	Sequences	No	Very low	Long-range dependencies
Bayesian (L10–12)	Any	Full posterior	High	Uncertainty, small samples

No single method dominates. Choose based on: (1) data type and size, (2) need for uncertainty quantification, (3) need for interpretability, (4) time to deploy. In practice, combine: use XGBoost for the point forecast and a Bayesian model for uncertainty intervals.

Lecture 12: Key Takeaways

1. **Bayesian linear regression** places priors on coefficients and returns a full posterior distribution for each β_j. A Normal prior is equivalent to Ridge regularization; a Laplace prior is equivalent to LASSO.
2. Posterior coefficient plots give: posterior mean, HDI, and $P(\beta_j > 0)$ — richer information than a single p-value.
3. **DAGs** make the causal structure explicit, helping identify confounders (must include) and colliders (must exclude). Always draw the DAG before specifying a regression model.
4. **Scenario analysis** with posterior samples: plug any counterfactual $\mathbf{x}^*$ into posterior draws to get a full predictive distribution for that scenario.
5. **No single method is best for all forecasting problems.** Build intuition for when to use each tool in this course's toolkit, and combine methods (e.g., point forecast + Bayesian uncertainty) when the decision stakes are high.

References I